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Alexander Alexiev

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Profile

I am a graduate student in Mechanical Engineering at MIT, advised by Prof. Sangbae Kim. My research focuses on proprioceptive control for dexterous manipulation, combining reinforcement learning, optimal control, tactile sensing, and custom robotic hardware. I work across the entire robot stack from high-level learning algorithms to actuator level firmware and electronics to enable high bandwidth control and accurate proprioceptive feedback for contact-rich manipulation. Previously, I studied Engineering Science at the University of Toronto and worked on robot learning and ingestible medical robotics for autonomous drug delivery and sensing.

Education

M.S. **Massachusetts Institute of Technology**, Mechanical Engineering (Robotics) Expected May 2027

- Prof. Sangbae Kim; researching robot learning and proprioceptive control for dexterous manipulation.
- GPA: 4.8/5.0

BASc **University of Toronto**, Engineering Science (Robotics and AI) Sept 2020 – May 2025

- Prof. Florian Shkurti; researching computer vision and manipulation of deformable objects.
- Prof. Giovanni Traverso; researching ingestible medical robotics for autonomous drug delivery and sensing.
- Major GPA: 3.96/4.0

Publications

See to Reach, Feel to Grasp: Learning A Blind Grasp Reflex for Anthropomorphic Robotic Hands Sept 2026
Alexander Alexiev, Tzu-Yuan Lin, Sang Min Kim, Ho Jae Lee, Yonghyeon Lee, Sangbae Kim
10.48550/arXiv.2609.31323 (Under review, ICRA 2027)

Real-Time Force Regulation for Whole-Hand Dexterous Grasping Sept 2026
Sang Min Kim, ***Alexander Alexiev***, Tzu-Yuan Lin, Sangbae Kim, Young Min Kim, Yonghyeon Lee
10.48550/arXiv.2609.30082 (Under review, ICRA 2027)

An ingestible robot for autonomously controlled oral delivery of biologics June 2026
J. Chen*, K. Werner*, ***Alexander Alexiev****, G. Vollherbst*, et al. *(* equal contribution; co-first authors)*
Under review, *Nature Mechanical Engineering*

Learning Reactive Dexterous Grasping via Hierarchical Task-Space RL Planning and Joint-Space QP Control May 2026
Ho Jae Lee, Yonghyeon Lee, ***Alexander Alexiev***, Tzu-Yuan Lin, Se Hwan Jeon, Sangbae Kim
arXiv:2605.03363 (Under review, *IEEE Transactions on Robotics*)

An Ingestible Capsule for Luminance-Based Diagnosis of Mesenteric Ischemia Oct 2025
Jack Chen*, ***Alexander Alexiev****, et al. *(* equal contribution; co-first authors)*
10.1126/scirobotics.adx1367 (***Science Robotics***)

Hierarchical Reactive Grasping via Task-Space Velocity Fields and Joint-Space Quadratic Programming Sept 2025
Yonghyeon Lee, Tzu-Yuan Lin, ***Alexander Alexiev***, Sangbae Kim
10.48550/arXiv.2509.01044 (**ICRA** 2026)

High-Bandwidth Tactile-Reactive Control for Grasp Adjustment Sept 2025
Yonghyeon Lee, Tzu-Yuan Lin, ***Alexander Alexiev***, Sangbae Kim
10.48550/arXiv.2509.15876 (**ICRA** 2026)

An Implantable System for Opioid Safety Aug 2024
Hen-Wei Huang, Peter R. Chai, et al., ***Alexander Alexiev***, et al., Giovanni Traverso
10.1016/j.device.2024.100517 (***Device***, Cell Press)

Patents

Electrically-Triggered Microactuator Powered by Aluminum-Nickel Nanofilm
Patent, WO 2026/076319, April 2026

Research Experience

Research Assistant in Biomimetic Robotics Laboratory, Massachusetts Institute of Technology — Boston, MA
June 2025 – present

- Investigating performance gains from quasi-direct-drive (QDD) actuators for sim-to-real transfer in reinforcement learning for dexterous manipulation tasks.
- Created a novel method to grasp objects blindly using only hand proprioception via reinforcement learning and model-based optimal control; *See to Reach, Feel to Grasp*, under review at ICRA 2027.
- Implemented model-based grasping methods using collision-aware QPs for task-space velocity control and tactile-reactive grasp adjustment on custom hardware, performing real-world experiments for two ICRA 2026 papers.
- Developed firmware and a machine-learning training pipeline for soft fingertip sensors that estimate force and contact location at 200 Hz using CAN and LCM.

Undergraduate Researcher in Robot Vision and Learning Lab, University of Toronto — Toronto, ON
June 2022 – Apr 2025

- Undergraduate thesis combining non-rigid point-set registration with a graph neural dynamics model for real-time state estimation and prediction of rope and cloth, robust to occlusion and self-overlap.
- Created an RGB-D computer-vision pipeline with a custom CNN to improve a graph-based task-and-motion-planning algorithm.

Robotics Research Co-op in Lab for Translational Engineering, Harvard Medical School / MIT — Boston, MA
June 2023 – Jan 2025

- Designed an ingestible robot to deliver mRNA vaccines into stomach tissue, including an ultra-low-power microheater array for igniting reactive metallic nanofilm and an actuator that converts the energy into a penetrating jet of mRNA; under review at *Nature Mechanical Engineering*.
- Created an ingestible robot with a 10-channel light-sensor array to autonomously diagnose ischemia and gastric bleeding; co-first author in *Science Robotics*.
- Developed an embedded C++ algorithm for an implantable device that detects opioid overdose from live IMU data and autonomously administers naloxone; validated detection in a swine terminal study; published in *Device* (Cell Press).

Industry Experience

Machine Learning Research Intern, Electronic Arts — Vancouver, BC (Remote)
May 2021 – Sept 2021

- Led development of a computer-vision model for detecting game-engine animation glitches and non-realistic behavior in FIFA and NHL titles.
- Designed an autoencoder CNN for anomaly detection on rendered gameplay assets, detecting cloth-animation glitches, missing textures, and irregular human-body configurations.

Student Teams

Software Co-Lead, RoboSoccer Humanoid Robotics Team — Sept 2022 – Sept 2023

- Ported bipedal motion planning and control from MATLAB to a custom STM32 implementation, increasing the feedback loop to 1,000 Hz; designed a decision-tree-based multi-agent controller for four humanoid robots at RoboCup 2022.

Embedded Systems and Controls Team, University of Toronto Formula Racing — Sept 2020 – Sept 2022

- Wrote ATmega 2560 firmware for I2C and CAN sensor data and vehicle subsystems including throttle and ABS; designed a cell-level fusing system for a custom 600 V Li-ion battery pack.

Community Involvement

FIRST Robotics Team 7558 — Sept 2018 – present

- Teach modern robotics methods and tools to high-school students. Currently developing a C++ localization library using NVIDIA Isaac Sim, ROS 2, multiple cameras, AprilTags, an extended Kalman filter, and factor graphs.

References

Dr. Sangbae Kim, Professor of Mechanical Engineering — Massachusetts Institute of Technology
Dr. Giovanni Traverso, Professor of Mechanical Engineering — Massachusetts Institute of Technology
Dr. Florian Shkurti, Professor of Computer Science — University of Toronto